Lesson plan · for the teacher

Decisions and debugging

Programming · Level 1 · Lesson 3

Time
40 min

Materials
Chalk for a grid with a wall; Arrow cards and paper; A counter (a bottle top) for the robot

Learning goals

- Write decisions as IF … THEN … ELSE, with AND and OR.
- Debug with a routine.
- Treat bugs as normal, for pupils and for you.

Activity: Simon says IF

1. Warm-up (5 min): “IF you have a sister, THEN stand up, ELSE touch your nose.” Then join conditions with AND and OR, and watch who gets it right.
2. Teach (10 min): Write the debugging routine on the board: test, find the first wrong step, change one thing, test again. Show it on a robot program with one bug.
3. Swap and debug (15 min): Pairs write a robot program on paper and hide one bug in it. They swap; the other pair runs it with a counter on the grid and finds the bug.
4. Share (5 min): Two pairs explain how they found the bug, step by step. Praise the routine, not the speed.
5. Wrap-up (5 min): Ask: who made a bug today? Everyone. Then give the exit ticket.

Make a mistake on purpose early in the lesson and debug it out loud: pupils copy how you treat bugs.

Discussion

- Why start from the first step that goes wrong?
- What is the difference between AND and OR in a school rule?
- How can we make bugs feel normal in our class?

Exit ticket

- Write one rule with AND and one with OR from our school day.

Try it in class tomorrow
Play “Simon says IF”: “IF you are wearing blue, THEN stand up, ELSE clap.” Add AND and OR. Then give a rule with a deliberate mistake and ask pupils to debug it.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **A pupil's robot crashes into a wall, and she wants to rewrite the whole program. What should she do first?**

◯ Rewrite the whole program ◯ Change all the turns

◯ Run it and find the first step that goes wrong

2 **The robot should go forward 2, turn left, forward 3, turn right, forward 1. Here is the program, one card per line. Which line is wrong?**

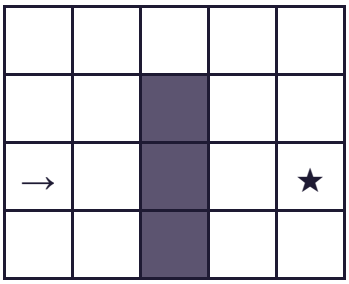
Circle the line with the mistake. Write it correctly:

```
1  forward(2)
2  right()
3  forward(3)
4  right()
5  forward(1)
```

3 **Rule: "Go out to play IF it is break time AND it is not raining." It is break time, and it is raining. What happens?**

◯ Half the class goes out ◯ Everyone stays in ◯ Everyone goes out

4 **A wall stands between the robot and the flag. Build a program that goes around it, with 7 cards at most.**



Cut out cards (↑ forward, ↶ turn left, ↷ turn right) and lay out a program that takes the robot to the ★. Write it here:

5 **Good programmers write programs with no bugs the first time.**

◯ True ◯ False

Answers · for the teacher

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1 **Run it and find the first step that goes wrong**

Everything before the first wrong step works, so start there: change one thing, then test again.

2 **Line 2: `left()`**

The first turn should be left. Everything before line 2 is right, so that is the first step that goes wrong: fix it, and test again before touching anything else.

3 **Everyone stays in**

With AND, both parts must be true. "It is not raining" is false, so the whole condition is false and nobody goes out.

4 **Any program that takes the robot to the ★ with 7 cards or fewer.**

Turn left, forward × 2, turn right, forward × 4, turn right, forward × 2: six cards over the top of the wall.

5 **False**

Every programmer writes bugs; good ones find and fix them with a routine.

✎ **For you to think about**

How do you react when your own plan goes wrong in class? How could you show pupils the debugging routine, out loud, next time it happens?